



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

FABRICATION ENGINEERING

SUBJECT: Engineering Drawing I

PAPER NO: 324/18/803A

DURATION: 3 hours

NOVEMBER/DECEMBER 2022 EXAMINATION

REQUIREMENTS

1. A3 drawing board
2. A3 drawing paper
3. Drawing instruments
4. Non-programmable scientific calculator

INSTRUCTIONS TO CANDIDATE

1. This paper contains Section A, B and C.
2. Answer all questions.

This paper consists of 7 printed pages.

SECTION A

QUESTION 1

Sketch the BS499 welding symbols for the following welds.

- | | | |
|----|--|-----------|
| a) | Double U-butt | (2 marks) |
| b) | Edge | (2 marks) |
| c) | Square butt | (2 marks) |
| d) | Seal | (2 marks) |
| e) | Spot weld (resistance welding) projection weld | (2 marks) |

QUESTION 2

- | | | |
|----|--|-----------|
| a) | Sketch the convention of: | |
| | i) First angle projection | (2 marks) |
| | ii) Third angle projection | (2 marks) |
| b) | Write down the meaning of the following abbreviations: | |
| | i) PCD | (1 mark) |
| | ii) CRS | (1 mark) |
| | iii) A/F | (1 mark) |
| | iv) R5J | (1 mark) |
| | v) CSK | (1 mark) |
| | vi) A/C | (1 mark) |

SECTION B

QUESTION 4

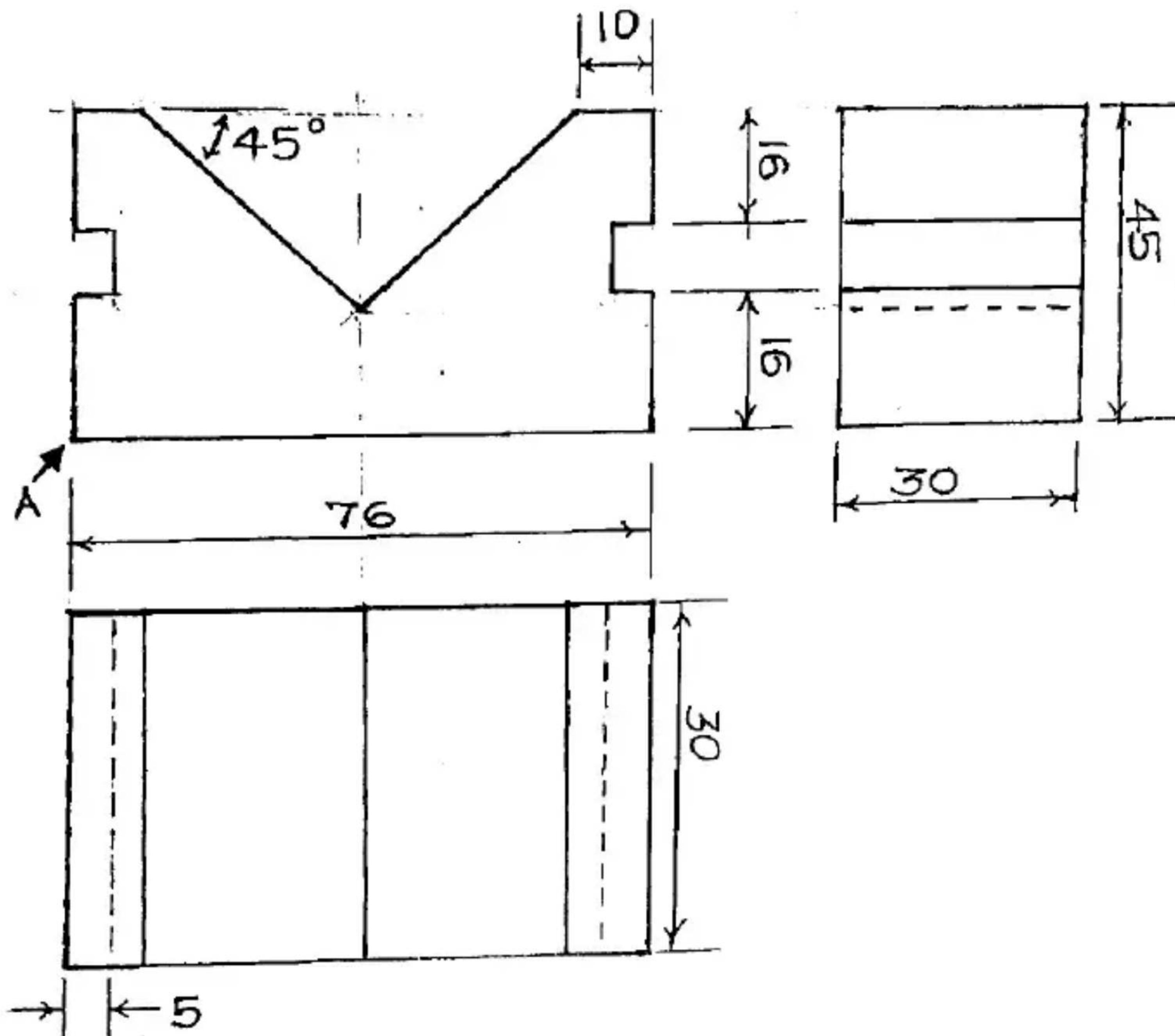


Fig. Q4

The drawing Fig. Q4 shows the elevations of a Vee block. Do not copy these views but draw an isometric view of the block showing corner A at the lowest point on the view. Hidden edges are not to be shown. Do not add dimensions. (15 marks)

QUESTION 5

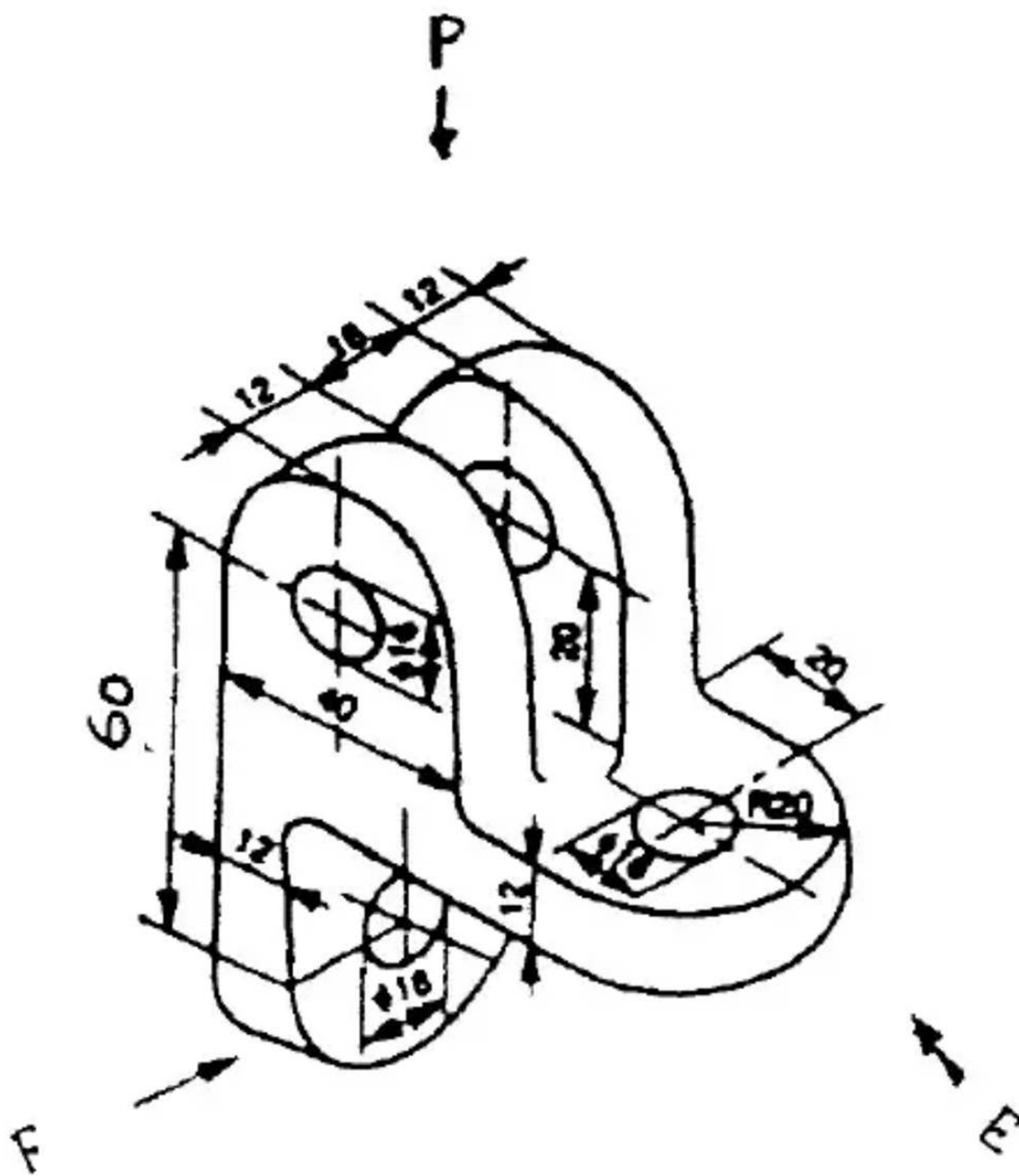


Fig. Q5

A pictorial view of a casting is shown in Fig. Q5. Draw in first angle projection, the following views.

- A front elevation viewed in the direction of arrow F.
- An end elevation viewed in the direction of arrow E.
- A plan view P.
- Radii of 3mm may be drawn freehand.

(15 marks)

SECTION C

QUESTION 6

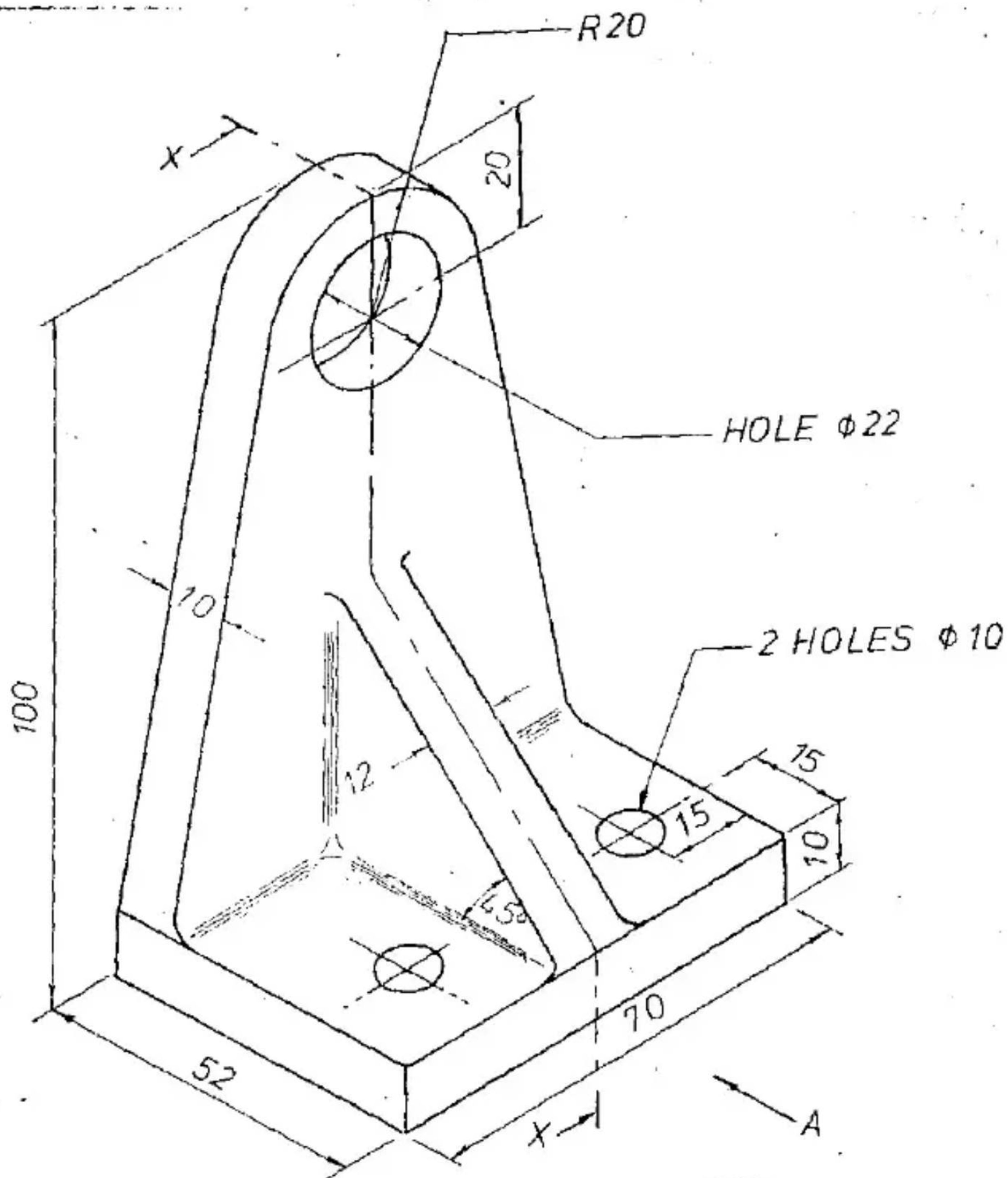


Fig. Q6

A bracket, Fig. Q6, is shown in isometric projection. Do not copy, but draw to a scale of full size the following views.

- A front elevation as seen from the direction of arrow A. (7 marks)
- A sectional view, the direction of the view and the cutting plane being shown by line X-X. (6 marks)

- c) A plan in projection with the front elevation.

(7 marks)

Do not add dimensions. Do not show any hidden details. All fillet radii are 3mm and can be drawn freehand.

QUESTION 7

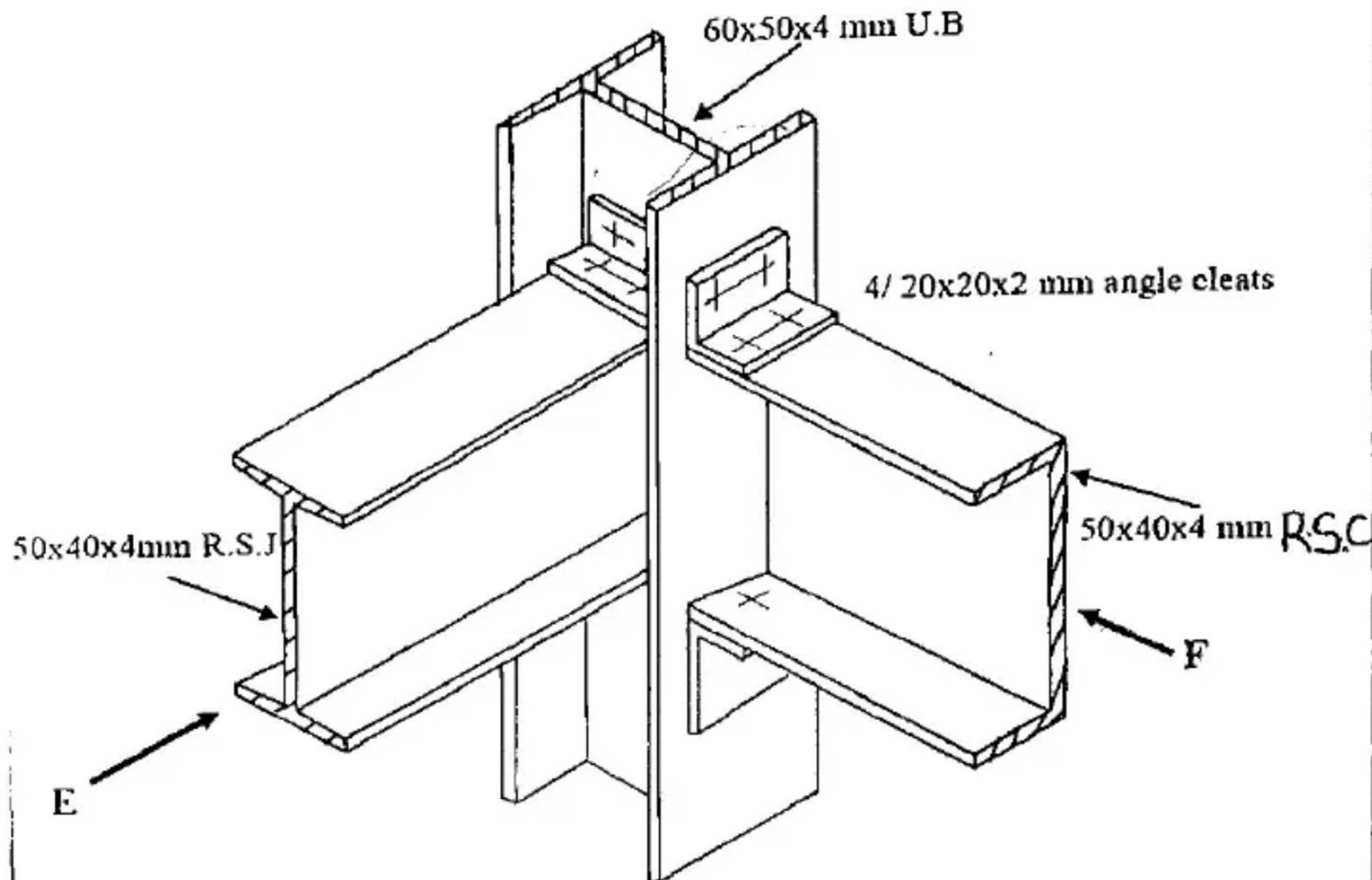


Fig. Q7

Fig. Q7 shows a stanchion connected centrally by two horizontal members using cleats. Do not copy this drawing, but draw the following:

- The front view as seen from the direction of arrow F. (7 marks)
- The end view as projected from the direction of arrow E. (7 marks)
- A plan in projection with the front view. (6 marks)

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